4G and 5G Mobile Network Architecture

An in-depth reference guide — radio, core, IMS, identity, QoS, mobility and the points where communications data is generated

Prepared for Michael Rossi
October 2026 — full edition

Contents

Contents **2**

Part I — Orientation **5**

1.1 Generations in one paragraph 5

1.2 The three planes 5

1.3 Reading a 3GPP diagram 5

Part II — 4G: the Evolved Packet System **6**

2.1 The picture 6

2.2 The access network: E-UTRAN 6

2.3 The core: EPC 6

2.4 Interfaces 7

2.5 Bearers and QoS 8

2.6 Procedures 8

2.7 CUPS: separating control and user plane in 4G 9

Part III — 5G: the 5G System **10**

3.1 The picture 10

3.2 The access network: NG-RAN 10

3.3 The core: 5GC and the service-based architecture 10

3.4 Reference points and services 11

3.5 NSA and SA deployment options 12

3.6 PDU sessions and QoS flows 12

3.7 Network slicing 13

3.8 Identity and privacy in 5G 13

3.9 Procedures 13

Part IV — The layers common to both **15**

4.1 IMS: the service core above the packet core 15

4.2 Identifiers across the layers 16

4.3 Charging 16

4.4 Lawful intercept architecture (3GPP TS 33.127) 17

4.5 Wi-Fi access 17

Part V — The CDA view **18**

5.1 Where each record is generated 18

5.2 Which node is authoritative 18

5.3 What the network does not generate 19

5.4 The 4G/5G migration traps 19

Appendix A — Interface index (4G and 5G side by side) **20**

Appendix B — Acronym glossary **21**

Appendix C — Specification index **23**

How to use this guide

This is the long-form reference for the mobile network architecture that a CDA design has to understand: the 4G Evolved Packet System (EPS) and the 5G System (5GS), the IMS service core that sits on top of both, and the identifiers, procedures and interfaces that produce communications data. It is organised so that each part can be read alone. Part I is orientation; Parts II and III are the 4G and 5G architectures in depth; Part IV covers what is common to both (IMS, identity, QoS, mobility, charging, lawful intercept); Part V is the CDA view. The abridged edition carries the same structure in a tenth of the space.

Part	Contents
I. Orientation	The generations, the three planes, how to read a 3GPP diagram
II. 4G / EPS	E-UTRAN, EPC, interfaces, bearers, attach and session procedures
III. 5G / 5GS	NR, the service-based core, NSA and SA, PDU sessions, slicing, procedures
IV. Common layers	IMS, identifiers, QoS, mobility and interworking, charging, lawful intercept
V. The CDA view	Where data is generated, which node is authoritative, what is not generated
Appendices	Interface index, acronym glossary, specification index

Part I — Orientation

1.1 Generations in one paragraph

2G and 3G carried voice as circuits and data as an add-on. 4G (LTE, standardised from 3GPP Release 8, 2008) is all-IP: there is no circuit domain, every service including voice is a packet flow, and the core network — the Evolved Packet Core — is a set of gateways and a control node. 5G (Release 15 onward, 2018) keeps the all-IP model but rebuilds the core as a set of cloud-native network functions talking over service-based interfaces, separates user plane from control plane completely, and adds network slicing and a new radio. Voice in both generations is delivered by the IMS, which sits above the core and is the same system for 4G, 5G, Wi-Fi calling and fixed broadband.

1.2 The three planes

Every element in this guide lives on one or more of three planes, and almost every architectural argument comes down to which plane a piece of data is on.

Plane	What travels on it	4G elements	5G elements
Control plane (signalling)	Attach, authentication, session set-up, mobility, policy decisions	MME, HSS, PCRF, the control side of SGW/PGW	AMF, SMF, UDM, AUSF, PCF, NRF, NSSF, NEF
User plane (traffic)	The subscriber's IP packets — web, video, and the voice media of an IMS call	SGW-U, PGW-U (after CUPS), eNB user plane	UPF, gNB user plane
Management plane	Configuration, performance, fault, orchestration	OSS/BSS, EMS	OSS/BSS, NFV MANO, SMO

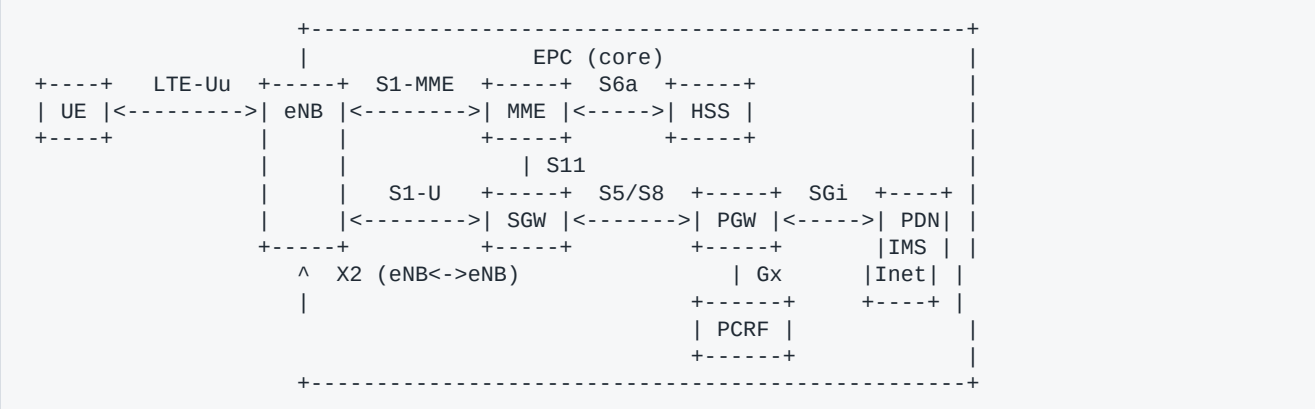
For communications data the control plane is what matters: it is where identities, times, cells and session events are visible. The user plane carries content and, after CGNAT, the address translations that resolution depends on. The management plane is where change happens, usually without telling anyone downstream.

1.3 Reading a 3GPP diagram

Boxes are functions, not necessarily machines: a vendor may combine several in one product (an HSS+UDM, an SGW/PGW combined node) or split one across many instances. Lines are reference points named with a letter and number (S1, S6a, N2, N11) and each carries a defined protocol. In 4G the reference points are point-to-point; in 5G the control-plane functions also expose services (Namf, Nsmf) on a shared bus, and the N-numbered reference points are a logical view over those services. Whenever a vendor says 'our platform does X', ask which 3GPP function and which reference point they mean.

Part II — 4G: the Evolved Packet System

2.1 The picture



2.2 The access network: E-UTRAN

E-UTRAN is the radio access network and consists of eNodeBs (eNBs) — the base stations — and nothing else. There is no controller node (the 3G RNC is gone); the eNB does radio resource management, scheduling, handover decisions, encryption of the radio link and admission control itself. eNBs connect to each other over X2 for handover and interference coordination, and to the core over S1.

Concept	Meaning
Cell	One sector of one eNB on one carrier frequency. Identified globally by E-UTRAN Cell Global Identifier (ECGI = PLMN + 28-bit cell identity). This is the 4G location granularity.
Tracking Area (TA)	A group of cells. The UE in idle mode is known to the core only at TA level (Tracking Area Identity, TAI); paging goes to all cells in the TA list.
RRC	Radio Resource Control — the protocol between UE and eNB that establishes the radio connection. RRC_IDLE and RRC_CONNECTED are the two states that matter.
Carrier aggregation, MIMO	Capacity features; irrelevant to CDA except that a UE may be served by several cells at once and the primary cell is the one reported.
eNB types	Macro, small cell, home eNB (femto). A femto cell's reported location is the device's registered address, which may be wrong.

2.3 The core: EPC

Element	Role	What it knows
MME — Mobility Management Entity	The control-plane brain. Handles attach, authentication (with HSS), security context, tracking-area updates, paging, bearer set-up signalling, handover control, and selection of SGW/PGW. One MME serves a pool of eNBs.	IMSI, IMEI (after identity request), current TAI and last-known ECGI at the moment of a signalling event, attach/detach times, bearers active, which SGW serves the UE
HSS — Home Subscriber Server	The subscriber database. Authentication vectors, subscription profile (APNs allowed, QoS, roaming), current serving MME. Shared with IMS (identity bindings).	IMSI to MSISDN, subscribed services, which MME the UE is registered on now

Element	Role	What it knows
SGW — Serving Gateway	User-plane anchor within the visited network. Routes packets between eNB and PGW; local mobility anchor during handover; buffers downlink packets for idle UEs.	Which eNB a UE's bearers currently terminate on; bearer-level volume counters
PGW — PDN Gateway	The edge of the mobile network towards the internet or IMS. Allocates the UE IP address, enforces policy and charging (PCEF), performs DPI where deployed, and is where NAT typically sits.	UE IP address allocation and release times, APN, data volumes, charging records, NAT bindings if co-located
PCRF — Policy and Charging Rules Function	Decides QoS and charging rules per flow. The IMS asks it (over Rx) to set up a dedicated voice bearer; it instructs the PGW (over Gx).	When a voice bearer was requested and released — corroboration for IMS session timing
Other	EIR (equipment identity register — IMEI blacklist); OCS/OFCS (online/offline charging); SMSC; ANDSF; ePDG (for untrusted Wi-Fi access to EPC); TWAG (trusted Wi-Fi).	

2.4 Interfaces

Reference point	Between	Protocol	Carries
LTE-Uu	UE and eNB	RRC / PDCP / RLC / MAC / PHY	The radio link
X2	eNB and eNB	X2AP over SCTP; GTP-U for data forwarding	Handover, interference coordination
S1-MME	eNB and MME	S1AP over SCTP (NAS carried inside)	Attach, bearer set-up, handover, paging; NAS messages between UE and MME transit here
S1-U	eNB and SGW	GTP-U over UDP	User-plane tunnels, one per bearer
S6a	MME and HSS	Diameter	Authentication vectors, location update, subscription data
S11	MME and SGW	GTP-C v2	Create/modify/delete session and bearer
S5 / S8	SGW and PGW (S8 when roaming)	GTP-C v2 and GTP-U (or PMIPv6)	Session and bearer management; user-plane tunnels
Gx	PGW (PCEF) and PCRF	Diameter	Policy and charging rules per flow
Rx	P-CSCF (IMS) and PCRF	Diameter	Request for a dedicated media bearer
SGi	PGW and external network	IP	The subscriber's traffic to the internet or to the IMS
S10	MME and MME	GTP-C v2	Inter-MME handover and context transfer
S13	MME and EIR	Diameter	IMEI check
Gy / Gz	PGW and OCS / OFCS	Diameter / GTP'	Online and offline charging
SWm / S2b	ePDG and AAA / PGW	Diameter / GTP	Untrusted Wi-Fi (Wi-Fi calling) access

2.5 Bearers and QoS

The EPS bearer is the unit of QoS. Every UE with an active PDN connection has at least one default bearer per APN, created at attach, which is non-guaranteed-bit-rate and carries general traffic. Dedicated bearers are added for flows needing specific treatment; the one that matters is the voice bearer the IMS requests for a call.

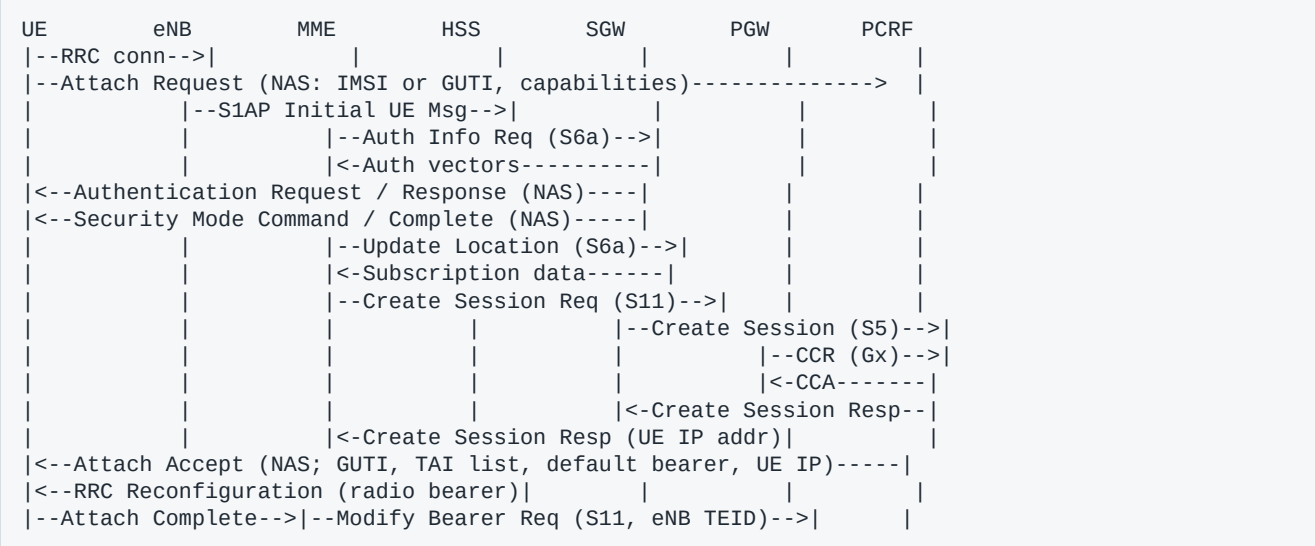
```
UE ===== radio bearer ===== eNB ==== S1 bearer ==== SGW ==== S5/S8 bearer ==== PGW
     <------------------------- EPS bearer (end to end) -------------------------->
     one EPS bearer = one QoS class = one GTP tunnel on each hop
```

QCI	Type	Typical use	CDA relevance
1	GBR	Conversational voice (VoLTE media)	A QCI-1 bearer set-up/release pair brackets a call's media
2	GBR	Conversational video	
5	Non-GBR, highest priority	IMS signalling (the 'ims' APN default bearer)	This bearer exists from attach for every VoLTE-capable device; its presence is registration, not a call
6-9	Non-GBR	Internet traffic, by subscriber tier	The 'internet' APN default bearer; where ICR-type data would be seen
65, 66, 69, 70	GBR / non-GBR	Mission-critical push-to-talk and signalling	

An APN (Access Point Name) names a PDN connection. VoLTE devices attach to at least two: the 'ims' APN (QCI 5 default bearer, IMS signalling) and the 'internet' APN. Each APN gets its own UE IP address, so a device has at least two addresses at once — which is why 'the UE's IP address' is never a complete identifier.

2.6 Procedures

Attach



What this produces: an HSS location update (which MME now serves this IMSI), an MME record with IMSI, IMEI, TAI and ECGI, a PGW address allocation with a timestamp, and a Gx session. Three of those are communications-data events; the IMS registration that follows is a fourth.

Tracking area update and paging

In idle, the UE moves freely within its TA list and reports only when it leaves it or a periodic timer fires. The core therefore knows an idle UE's location only at TA granularity and only as at the last update. When traffic arrives for an idle UE, the MME pages across the TA list; the UE answers from one cell and that cell is then known. This is the mechanism behind 'location is occasional, not continuous'.

Dedicated bearer for a VoLTE call

```
P-CSCF --AAR (Rx: media description)--> PCRF --RAR (Gx: new rule)--> PGW
PGW --Create Bearer Request (S5)--> SGW --Create Bearer Request (S11)--> MME
MME --E-RAB Setup (S1AP)--> eNB --RRC Reconfiguration--> UE
... call ...
P-CSCF --STR (Rx)--> PCRF --RAR--> PGW --Delete Bearer--> ... --> UE
```

Handover

Type	Path	Core involvement
X2 handover	eNB to eNB directly, same MME and SGW	MME told afterwards (Path Switch); SGW re-points the S1-U tunnel
S1 handover	Via the MME, when X2 is absent or MME/SGW changes	MME (and possibly a second MME over S10) orchestrates
Inter-RAT to 3G/2G	SRVCC for a call in progress; PS handover for data	MME to MSC server (Sv) for SRVCC
Inter-system to 5G	N26 between MME and AMF	Context transferred; session continues on 5GC

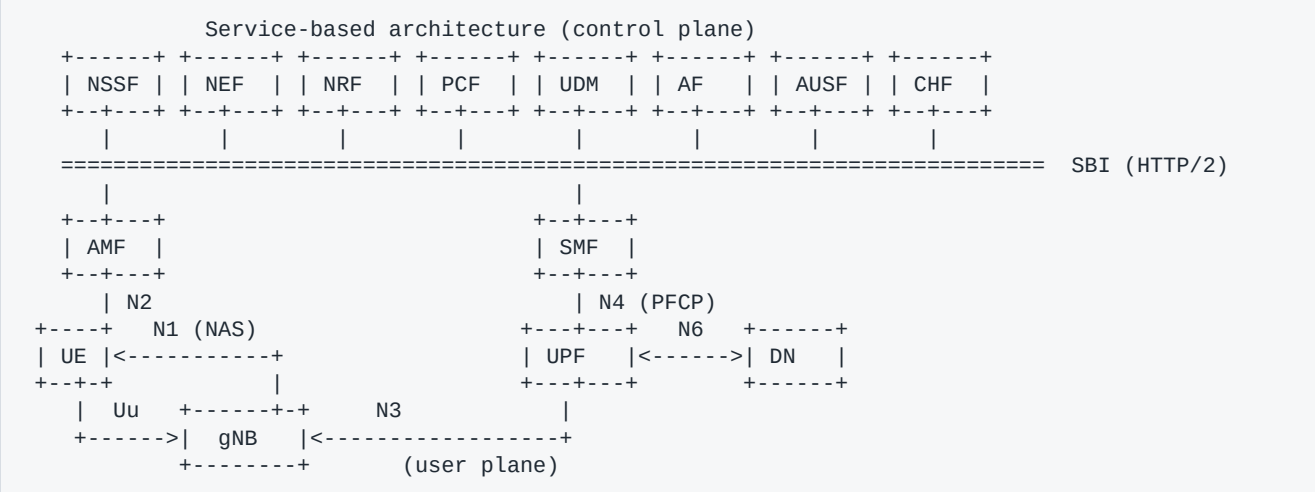
Every handover updates the serving cell known to the eNB, but only S1 handovers and path switches are visible to the core as events. Cell changes within a connected session are therefore under-reported in core records; the P-Access-Network-Info header on the next IMS signalling message is often the better location source.

2.7 CUPS: separating control and user plane in 4G

Release 14 introduced Control and User Plane Separation: SGW and PGW each split into a control part (SGW-C, PGW-C) and a user part (SGW-U, PGW-U) linked by Sx (PFCP). This lets user-plane nodes be placed near the edge and scaled independently. It is the direct ancestor of 5G's SMF/UPF split, and in mixed estates the same physical UPF often serves as PGW-U.

Part III — 5G: the 5G System

3.1 The picture



3.2 The access network: NG-RAN

Concept	Meaning
gNB	The 5G base station, serving NR cells. Internally split into a Central Unit (CU, handling RRC/PDCP) and Distributed Units (DU, handling RLC/MAC/PHY) over F1; the CU itself may split into CU-CP and CU-UP over E1. Open RAN (O-RAN) standardises these splits across vendors.
ng-eNB	An LTE base station connected to the 5G core (NG interface) — used in some SA migration paths.
NR Cell Global Identity (NCGI)	PLMN + 36-bit NR cell identity. The 5G location granularity; the cell identity is longer than 4G's and is a different identifier space.
Registration Area	The 5G equivalent of the tracking-area list; the AMF knows an idle UE at RA granularity.
Frequency ranges	FR1 (sub-6 GHz, the practical bulk of coverage) and FR2 (mmWave, dense urban and venues). mmWave cells are tiny, so cell identity is unusually precise location in FR2.
Beamforming, massive MIMO	Capacity and coverage features; no CDA consequence except that beam-level measurements are not retained as location.

3.3 The core: 5GC and the service-based architecture

The 5G core is designed cloud-native. Every control-plane function is a network function (NF) that offers services over HTTP/2 REST APIs on a common bus — the Service-Based Interface (SBI). Functions find each other through the NRF rather than through configured point-to-point links. The 4G MME is split into the AMF (mobility, access, NAS security) and the SMF (sessions); the SGW/PGW split into SMF (control) and UPF (user plane); the HSS becomes UDM plus UDR (data) plus AUSF (authentication); the PCRF becomes the PCF.

NF	Role	4G ancestor	What it knows (CDA view)
AMF — Access and Mobility Management	Registration, NAS signalling termination, security, mobility, reachability, paging; selects SMF	MME (mobility half)	SUPI/PEI, registration area, last-known NCGI at signalling events, registration/deregistration times, which SMFs hold sessions
SMF — Session Management	PDU session establishment/modification/release; IP address allocation; selects and controls UPFs over N4; charging triggers	MME (session half) + SGW-C/ PGW-C	UE IP address allocation and release, DNN, session times, which UPF anchored the session
UPF — User Plane Function	Packet routing and forwarding, QoS enforcement, usage reporting, DPI, NAT, lawful-intercept user-plane POI, anchor for mobility	SGW-U/PGW-U	Usage volumes; NAT translations; the user-plane content path (out of CDA scope)
UDM — Unified Data Management	Subscription management, authentication credential generation (with AUSF), SUPI/SUCI de-concealment, serving-NF registration	HSS (front end)	SUPI to MSISDN (GPSI), subscribed services and slices, which AMF serves the UE now
UDR — Unified Data Repository	The storage behind UDM, PCF and NEF	HSS (database)	The persistent subscription record
AUSF — Authentication Server	Runs 5G-AKA or EAP-AKA' with the UE via AMF	HSS (auth vectors)	Authentication success/failure events
PCF — Policy Control	QoS, charging and access policy per session and per flow; the Rx/N5 counterpart for IMS	PCRF	When a voice QoS flow was requested and released
NRF — Network Repository	Service discovery and registration for all NFs	(DNS/config)	Which NF instances existed — change evidence
NSSF — Network Slice Selection	Decides which slice(s) a UE may use and which AMF serves them	(none)	Slice assignment per UE
NEF — Network Exposure	Secure exposure of network capabilities to external applications; the API gateway	SCEF	Third-party interactions with subscriber context
CHF — Charging Function	Converged online/offline charging over Nchf	OCS/OFCS	The charging record
SMSF, SEPP, SCP, N3IWF	SMS over NAS; roaming security edge proxy; service communication proxy; untrusted non-3GPP (Wi-Fi) interworking	SMSC, (IPX), (none), ePDG	SMS over NAS events; roaming partner identity; Wi-Fi access events

3.4 Reference points and services

Reference point	Between	Protocol	Carries
Uu	UE and gNB	NR RRC / PDCP / RLC / MAC / PHY	The radio link
N1	UE and AMF	5G NAS (over N2)	Registration, authentication, PDU session requests, SMS over NAS
N2	gNB and AMF	NGAP over SCTP	UE context, PDU session resource set-up, handover, paging
N3	gNB and UPF	GTP-U	User-plane tunnels per PDU session

Reference point	Between	Protocol	Carries
N4	SMF and UPF	PFCP	Packet detection and forwarding rules, usage reporting, LI tasking of the UPF via the SMF
N6	UPF and data network	IP	Subscriber traffic to internet or IMS
N9	UPF and UPF	GTP-U	Chained user-plane functions
N26	AMF and MME	GTP-C v2	4G/5G interworking with session continuity
Xn	gNB and gNB	XnAP	Handover, dual connectivity
Namf, Nsmf, Nudm, Nausf, Npcf, Nnrf, Nnssf, Nnef, Nchf	Services offered by each NF on the SBI	HTTP/2 + JSON (OpenAPI)	Everything the control plane does; N8, N10, N11, N12, N15 etc. are logical names for pairs of these services
N5 (Npcf)	IMS P-CSCF (as AF) and PCF	HTTP/2	Media QoS flow request — the 5G Rx

3.5 NSA and SA deployment options

'5G' on a phone screen can mean two very different networks.

Option	Radio	Core	What it means for records
Option 3 / 3x (NSA, EN-DC)	LTE eNB is the anchor (control plane); NR gNB adds user-plane capacity as a secondary node	EPC (4G core)	There is no 5G core. All signalling, attach, bearers, IMS registration and session records are 4G records. The only 5G trace is a secondary-cell indication in RAN measurements.
Option 2 (SA)	NR gNB alone	5GC	True 5G: registration with AMF, PDU sessions via SMF/UPF, NCGI location, SUPI/SUCI identity, VoNR or EPS Fallback for voice
Option 4 / 7 (NR-DC, NE-DC)	Mixed anchors on 5GC	5GC	Rare in practice

> **GOVERNANCE NOTE**
>
> Most UK operators have run NSA for years and are moving to SA progressively. A retention design must therefore expect 'a 5G subscriber' to produce 4G-shaped records for some time, and both shapes concurrently during migration. Ask the programme which option each site, and each device class, is on.

3.6 PDU sessions and QoS flows

The 5G equivalent of an EPS bearer plus PDN connection is the PDU session: a connection to a Data Network Name (DNN — the 5G APN) with one or more QoS flows inside it, each identified by a QoS Flow Identifier (QFI) and governed by a 5QI value. Unlike 4G, one PDU session carries many flows in a single N3 tunnel; QoS differentiation is per flow, not per tunnel. PDU session types: IPv4, IPv6, IPv4v6, Ethernet, Unstructured.

5QI	Equivalent QCI	Use
1	1	Conversational voice (VoNR media)

5QI	Equivalent QCI	Use
2	2	Conversational video
5	5	IMS signalling — the 'ims' DNN default flow
6-9	6-9	Internet traffic tiers
80-86	(new)	Low-latency and V2X; delay-critical GBR

Session and Service Continuity (SSC) modes 1, 2 and 3 define whether the UE keeps its IP address when the anchor UPF changes. Mode 1 (keep) is the 4G behaviour; modes 2 and 3 release and re-allocate, which means a 5G UE's IP address may change mid-session in ways 4G never did — a new wrinkle for IP-to-subscriber resolution.

3.7 Network slicing

A slice is a logical end-to-end network — its own set of NF instances, UPFs and RAN resources — identified by S-NSSAI (slice/service type plus optional differentiator). Typical SSTs: eMBB (1, broadband), URLLC (2, low latency), mIoT (3, massive IoT), V2X (4). A UE may be registered to several slices at once; each PDU session belongs to exactly one. For CDA the consequences are that a subscriber's sessions may be anchored on different SMF/UPF sets depending on slice, that slice membership is itself a subscription attribute (entity data), and that a sliced IoT population will behave nothing like handsets in volume.

3.8 Identity and privacy in 5G

5G fixed the long-standing 4G weakness that the IMSI is sent in clear on first attach. The permanent identifier is now the SUPI (Subscription Permanent Identifier — in practice an IMSI or a network access identifier), which is never sent over the air; the UE sends a SUCI (Subscription Concealed Identifier), an encryption of the SUPI under the home network's public key, which only the home UDM can decrypt. The temporary identifier is the 5G-GUTI, re-allocated frequently. The device identity is the PEI (Permanent Equipment Identifier — the IMEI). The externally visible subscriber identity is the GPSI (Generic Public Subscription Identifier — the MSISDN or an external ID).

4G	5G	Note
IMSI	SUPI	Same value in practice for SIM-based subscriptions; never in clear on the air in 5G
(none)	SUCI	Concealed SUPI; resolved only by the home UDM
GUTI	5G-GUTI	Temporary; more frequent re-allocation
IMEI	PEI	Device
MSISDN	GPSI	Public subscriber identity; may also be an external identifier
IMSI-catcher exposure	Largely closed	Radio-side passive identification of subscribers is much harder in SA 5G

3.9 Procedures

Registration

```
UE        gNB        AMF        AUSF       UDM        PCF
|--RRC-->|          |          |          |          |
|--Registration Request (NAS: SUCI or 5G-GUTI, requested NSSAI)-->|
|           |--NGAP Initial UE Msg-->|   |          |
|           |          |--Nausf_UEAuth-->|          |
|           |          |          |--Nudm_UEAuth (de-conceal SUCI, vectors)-->|
|<--Authentication Request / Response (5G-AKA)-----|
|<--Security Mode Command / Complete--------------|
|           |          |--Nudm_UECM_Registration (serving AMF)-->|
|           |          |--Nudm_SDM_Get (subscription, slices)--->|
|           |          |--Npcf_AMPolicyControl_Create----------->|
|<--Registration Accept (5G-GUTI, allowed NSSAI, registration area)|
|--Registration Complete-->|
```

Note that in 5G, registration does not create a data session: PDU session establishment is a separate, later procedure (unlike 4G attach, which creates the default bearer). A registered-but-sessionless 5G UE is a normal state.

PDU session establishment

```
UE --PDU Session Establishment Request (NAS: DNN, S-NSSAI, type)--> AMF
AMF --Nsmf_PDUSession_CreateSMContext--> SMF
SMF --Nudm_SDM_Get (session subscription)--> UDM
SMF --Npcf_SMPolicyControl_Create--> PCF
SMF --N4 Session Establishment (PFCP: rules, UE IP)--> UPF
SMF --Namf_Communication_N1N2MessageTransfer--> AMF
AMF --NGAP PDU Session Resource Setup (N3 tunnel info)--> gNB
gNB --RRC Reconfiguration--> UE   ;  UE <-- NAS PDU Session Accept (IP addr, QoS rules)
gNB --NGAP Resource Setup Response--> AMF --Nsmf_UpdateSMContext--> SMF --N4 Modify--> UPF
```

Voice on 5G: VoNR and EPS Fallback

VoNR (Voice over New Radio) is an IMS call with its media on a 5QI-1 QoS flow in a PDU session to the 'ims' DNN — the same IMS, the same P-CSCF, a different access. Where VoNR is not yet enabled on a cell, the network uses EPS Fallback: on receiving the IMS call set-up, the gNB redirects or hands the UE over to LTE before the QCI-1 bearer is created, and the call completes on 4G. The IMS registration and the INVITE were on 5G; the dedicated bearer, the media and the release are on 4G. A single call therefore spans two cores and two record sets.

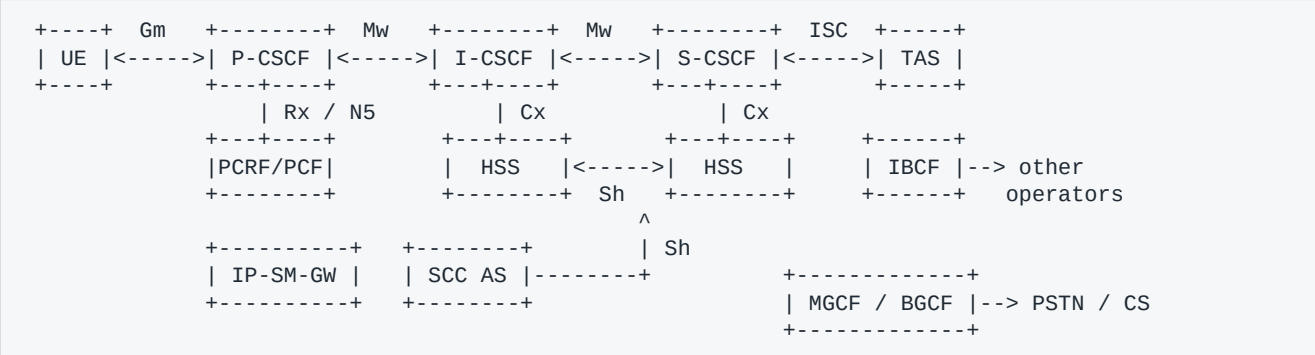
Mobility

Mechanism	Equivalent in 4G	Note
Xn handover	X2 handover	gNB to gNB; AMF told by Path Switch; UPF re-pointed by SMF
N2 handover	S1 handover	Via AMF, possibly across AMF sets
Inter-system via N26	S10 / inter-system	Session continuity to/from EPC; SUPI mapped to IMSI, 5G-GUTI to GUTI
Registration area update	Tracking area update	Idle-mode mobility
RRC_INACTIVE	(none)	A third RRC state: the gNB keeps the UE context and the UE can move within a RAN notification area without core signalling — even less core-visible location than 4G idle

Part IV — The layers common to both

4.1 IMS: the service core above the packet core

The IP Multimedia Subsystem is access-independent: the same core, the same identities and the same records serve a call made over 4G, 5G SA, Wi-Fi calling (via ePDG or N3IWF) or fixed broadband. It connects to the packet core in exactly two places: the P-CSCF reaches the UE through the 'ims' APN/DNN, and the P-CSCF asks the PCRF/PCF (Rx/N5) for a media bearer or QoS flow. Everything else about IMS is independent of the generation.



Node	Role	Interfaces
P-CSCF	First contact; access edge; asserts identity; requests QoS; compresses signalling; sees UE IP:port, access type and cell via P-Access-Network-Info	Gm (UE), Mw (CSCF), Rx/N5 (policy)
I-CSCF	Front door to the home network; asks HSS which S-CSCF serves the user	Mw, Cx
S-CSCF	Registrar and session controller; authenticates; holds the user profile; evaluates initial filter criteria and triggers application servers; routes	Mw, Cx, ISC
HSS (IMS part)	IMPI/IMPU bindings, service profile, S-CSCF assignment; shared with the EPC HSS and 5G UDM	Cx, Sh
TAS / MMTel AS	Supplementary services: diversion, barring, hold, transfer, conference, caller-ID presentation	ISC, Sh, Ut
SCC AS	Service centralisation and continuity: domain selection, SRVCC, access transfer	ISC, Sh
IP-SM-GW	SMS over IP	ISC, Sh, towards SMSC
IBCF / TrGW	Interconnect border; topology hiding; inserts inter-operator identifiers	Ici / Izi
BGCF / MGCF / IM-MGW	Breakout to circuit-switched networks and PSTN	Mj / Mg / Mn
MRF	Announcements, tones, conferencing media	Mr / Mp
E-CSCF / LRF	Emergency call routing and location	Mw, Ml

A VoLTE/VoNR call end to end

```
Caller UE   P-CSCF   S-CSCF(orig)  TAS   S-CSCF(term)  P-CSCF   Callee UE
|--INVITE-->|         |            |       |            |          |
|           |--INVITE (P-Asserted-Identity, P-Access-Network-Info, ICID)->|
|           |         |--iFC-->TAS |       |            |          |
|           |         |<-INVITE----|       |            |          |
|           |         |------- I-CSCF -> HSS -> S-CSCF(term) ----->|
|           |         |            |       |--iFC-->TAS |          |
|           |         |            |       |            |--INVITE->|
|<--183 Session Progress (SDP answer)-------------------------------|
|  P-CSCF --Rx/N5 AAR--> PCRF/PCF --> dedicated bearer / QoS flow (QCI/5QI 1)
|<--180 Ringing-----------------------------------------------------|
|<--200 OK (answered)-----------------------------------------------|
|--ACK--------------------------------------------------------------->|
|<================ RTP media, user plane, NOT through IMS ===========>|
|--BYE--------------------------------------------------------------->|
|<--200 OK----------------------------------------------------------|
|  P-CSCF --Rx/N5 STR--> release bearer / flow
```

4.2 Identifiers across the layers

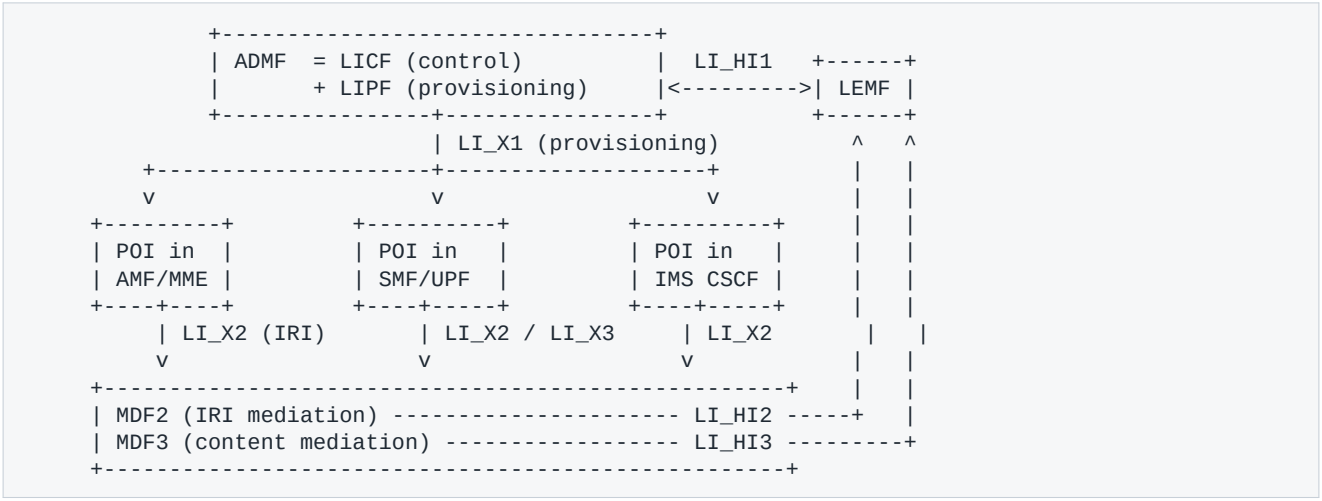
Identifier	Layer	Form	Stability	Use
IMSI / SUPI	Core subscription	MCC+MNC+MSIN (15 digits)	Per SIM	Ties core and IMS records to one subscription
IMEI / PEI	Device	15-16 digits	Per handset	What device was used; may be cloned
MSISDN / GPSI	Public	E.164	Portable, reassignable	The usual request starting point
GUTI / 5G-GUTI	Core, temporary	PLMN + MME/AMF id + TMSI	Changes often	Appears in signalling traces, not in disclosures
IMPI	IMS private	user@realm	Per subscription/ISIM	Authentication only
IMPU	IMS public	sip: or tel: URI	Several per subscriber	Throughout IMS session records
UE IP address	Session	IPv4 / IPv6	Per PDN/PDU session; shared via CGNAT	IP-to-subscriber resolution, with port and time
ECGI / NCGI	Location	PLMN + cell id (28 / 36 bits)	Per cell	Location at signalling moments
TAI / RA	Location, coarse	PLMN + TAC	Per area	Idle-mode whereabouts
ICID	Charging correlation	icid-value in P-Charging-Vector	Per session	Joins the records of one call across nodes
Charging ID	Core charging	Per bearer / PDU session	Per session	Joins core charging records
IOI	Interconnect	orig-ioi / term-ioi	Per leg	Names the peer operator
TEID	Tunnel	32-bit per GTP tunnel endpoint	Per bearer	Trace-level only

4.3 Charging

Both cores produce charging records: the 4G PGW generates per-bearer records over Gz (offline) and Gy (online); the 5G SMF reports to the CHF over Nchf. IMS nodes generate their own over Rf/Ro (4G) or Nchf

(5G), correlated by ICID. The charging record is the most complete pre-existing consolidated view of a session, which is why programmes are tempted to retain it — and why the CDA course warns that it reflects what is billable, changes with the tariff, and is consolidated late.

4.4 Lawful intercept architecture (3GPP TS 33.127)



Points of Interception (POIs) exist in the AMF, SMF, UPF, UDM, SMSF and in the IMS CSCFs and application servers. LI_X2 carries intercept-related information (events — the communications-data-shaped part); LI_X3 carries content. The same architecture is reused for communications data acquisition, which is the course's point that separation between content and CD is administrative.

4.5 Wi-Fi access

Wi-Fi calling reaches the same IMS through an IPsec tunnel from the UE to an ePDG (4G, via S2b to the PGW) or N3IWF (5G, via N2/N3 to AMF/UPF). The access-network information in IMS signalling then reports the access type as WLAN and may carry the UE's local IP address rather than a cell. Location is therefore absent or unreliable for Wi-Fi-originated sessions, and the public IP seen by the ePDG is the user's broadband address, not a mobile one.

Part V — The CDA view

5.1 Where each record is generated

Event	4G source	5G source	Also visible at
Subscriber attaches / registers to the network	MME (attach), HSS (location update)	AMF (registration), UDM (UECM registration)	PGW/SMF if a session is created
Device identity seen	MME (identity request, S13)	AMF (PEI), 5G-EIR	IMS registration (Contact header, +sip.instance)
Location at a signalling moment	MME (ECGI/TAI in S1AP), eNB	AMF (NCGI/TAI in NGAP), gNB	IMS P-CSCF via P-Access-Network-Info
IP address allocated / released	PGW (session create/delete)	SMF (PDU session)	Gx/Npcf session; RADIUS/DHCP if used
NAT translation	PGW-U / separate CGNAT	UPF / separate CGNAT	(nowhere else)
IMS registration	P-CSCF, S-CSCF, HSS	same	TAS (third-party registration)
Voice/video session set-up, answer, release	S-CSCF (authoritative), P-CSCF, TAS	same	PCRF/PCF bearer events; charging
Supplementary service applied	TAS	same	S-CSCF (via ISC)
Interconnect leg	IBCF	same	Charging (IOI)
SMS	SMSC / IP-SM-GW (over IMS) / MME (SMS over NAS, SGs)	SMSF (SMS over NAS) / IP-SM-GW	HSS/UDM for routing
Data session usage	PGW (charging)	SMF/UPF (usage reports), CHF	
Access change (4G/5G/Wi-Fi)	MME/SGW handover, SCC AS	AMF/SMF via N26, SCC AS	P-CSCF (access type changes in re-registration)

5.2 Which node is authoritative

Question	Authoritative source	Why
Who holds this number / which subscription?	HSS / UDM plus the business CRM	The binding is a subscription attribute, not a network event
Did a call happen, between whom, when?	S-CSCF	The only node that sees both identities and the final outcome on every leg
Where was the device at that moment?	P-CSCF (P-Access-Network-Info) for IMS events; MME/AMF for core events	The access-network identity is carried in the signalling
Which IP address did the subscriber have?	PGW / SMF	The allocator
Who was behind this public IP:port at this instant?	CGNAT logs	Nothing else holds the translation
Was the call diverted / barred / conferenced?	TAS	Service logic lives there

Question	Authoritative source	Why
Which operator was the other party on?	IBCF (IOI)	Inserted at the border
Was the device on 4G or 5G?	P-Access-Network-Info; AMF/MME context	Access type is an explicit field

5.3 What the network does not generate

- Continuous location: only cell identity at signalling moments; RRC_INACTIVE in 5G reduces it further.
- Content: media is on the user plane and never passes the IMS core; retained records do not and must not contain it.
- OTT services: a PDU session to a content-delivery network, with the destination increasingly hidden by encrypted DNS, encrypted SNI and QUIC.
- Who was holding the device.
- Reliable device identity per session: IMEI/PEI is captured at attach and at IMS registration, not per call.
- Reliable location for Wi-Fi-originated sessions.

5.4 The 4G/5G migration traps

- **NSA is 4G.** A 5G-labelled subscriber on Option 3x produces 4G core and 4G IMS records. No AMF, no SUPI, no NCGI.
- **EPS Fallback splits a call.** Registration and INVITE on 5G, bearer and media and release on 4G. Correlate on ICID, not on access type.
- **Cell identity spaces differ.** ECGI (28-bit) and NCGI (36-bit) are different identifiers; a cell-site reference table must hold both.
- **IP address behaviour differs.** SSC modes 2 and 3 can change the UE address mid-session; multiple PDU sessions per slice multiply addresses per device.
- **Identity concealment.** SUCI never appears in clear; records carry SUPI only after UDM de-concealment, so radio-side traces cannot be joined to subscribers without the core.
- **Separate charging functions.** OCS/OFCS and CHF may run in parallel with different record formats for years.
- **Feed inventory doubles.** Every 4G feed has a 5G counterpart; volume reconciliation must be per feed per generation.

> **GOVERNANCE NOTE**
>
> Ask the programme for a per-event matrix like 5.1, populated with the operator's actual nodes and vendors, with a nominated system of record per row and the deployment option (NSA/SA) per site. That matrix is the data supply agreement in embryo.

Appendix A — Interface index (4G and 5G side by side)

Function	4G reference point	5G equivalent
Radio	LTE-Uu	Uu (NR)
RAN to core control	S1-MME (S1AP)	N2 (NGAP)
RAN to core user plane	S1-U (GTP-U)	N3 (GTP-U)
UE to core NAS	NAS over S1-MME	N1 (NAS over N2)
Mobility node to subscriber DB	S6a (Diameter)	Nudm / Nausf (HTTP/2); logical N8, N12, N13
Mobility node to session node	S11 (GTP-C)	Nsmf / Namf; logical N11
Session control to user plane	Sx (PFCP, after CUPS)	N4 (PFCP)
Serving to PDN gateway	S5 / S8 (GTP)	N9 (UPF to UPF)
Policy	Gx (Diameter)	Npcf; logical N7
IMS to policy	Rx (Diameter)	N5 (HTTP/2)
Charging	Gy / Gz	Nchf (N40)
External network	SGi	N6
Inter-RAN	X2	Xn
Inter-mobility-node	S10	N14 (Namf)
4G/5G interworking	N26	N26
Equipment identity	S13 (EIR)	N17 (5G-EIR)
Untrusted Wi-Fi	SWu / S2b (ePDG)	NWu / N2, N3 (N3IWF)
Exposure	T8 (SCEF)	Nnef
Lawful intercept	X1 / X2 / X3	LI_X1 / LI_X2 / LI_X3

Appendix B — Acronym glossary

Acronym	Expansion
AMF	Access and Mobility Management Function (5G)
APN / DNN	Access Point Name (4G) / Data Network Name (5G)
AUSF	Authentication Server Function (5G)
CHF	Charging Function (5G)
CGNAT	Carrier-grade Network Address Translation
CSCF (P-, I-, S-, E-)	Call Session Control Function: Proxy, Interrogating, Serving, Emergency (IMS)
CUPS	Control and User Plane Separation (4G Rel-14)
ECGI / NCGI	E-UTRAN / NR Cell Global Identifier
EN-DC	E-UTRA–NR Dual Connectivity (NSA Option 3)
eNB / gNB	4G / 5G base station
EPC / EPS	Evolved Packet Core / Evolved Packet System
ePDG / N3IWF	Evolved Packet Data Gateway / Non-3GPP Interworking Function (Wi-Fi access)
GTP-C / GTP-U	GPRS Tunnelling Protocol, control / user
GUTI	Globally Unique Temporary Identifier
HSS / UDM / UDR	Home Subscriber Server / Unified Data Management / Unified Data Repository
IBCF	Interconnection Border Control Function
ICID	IMS Charging Identifier
IMPI / IMPU	IMS Private / Public User Identity
IMS	IP Multimedia Subsystem
IMSI / SUPI / SUCI	International Mobile Subscriber Identity / Subscription Permanent / Concealed Identifier
IMEI / PEI	International Mobile Equipment Identity / Permanent Equipment Identifier
IOI	Inter-Operator Identifier
LEMF	Law Enforcement Monitoring Facility
MME	Mobility Management Entity (4G)
MSISDN / GPSI	Mobile Subscriber ISDN Number / Generic Public Subscription Identifier
NAS	Non-Access Stratum (UE to core signalling)
NEF / NRF / NSSF	Network Exposure / Repository / Slice Selection Function
NSA / SA	Non-Standalone / Standalone 5G
PCRF / PCF	Policy and Charging Rules Function / Policy Control Function
PDU session	Protocol Data Unit session (5G data connection)
PFCP	Packet Forwarding Control Protocol (Sx / N4)
PGW / SGW	PDN Gateway / Serving Gateway (4G)
POI / MDF / ADMF	Point of Interception / Mediation and Delivery Function / Administration Function (LI)
QCI / 5QI / QFI	QoS Class Identifier / 5G QoS Identifier / QoS Flow Identifier

Acronym	Expansion
RRC	Radio Resource Control
SBA / SBI	Service-Based Architecture / Interface
SMF / UPF	Session Management Function / User Plane Function (5G)
S-NSSAI	Single Network Slice Selection Assistance Information
SRVCC	Single Radio Voice Call Continuity
SSC	Session and Service Continuity mode
TAS	Telephony Application Server
TAI / TAC	Tracking Area Identity / Code
TEID	Tunnel Endpoint Identifier
VoLTE / VoNR	Voice over LTE / Voice over New Radio

Appendix C — Specification index

Specification	Subject
3GPP TS 23.401	EPS architecture and procedures (4G core)
3GPP TS 23.501 / 23.502	5G System architecture / procedures
3GPP TS 23.503	5G policy and charging control framework
3GPP TS 23.228 / 24.229	IMS architecture / SIP profile
3GPP TS 23.003	Numbering, addressing and identification
3GPP TS 36.300 / 38.300	E-UTRAN / NG-RAN overall description
3GPP TS 36.413 / 38.413	S1AP / NGAP
3GPP TS 24.301 / 24.501	4G NAS / 5G NAS
3GPP TS 29.244	PFCP (Sx / N4)
3GPP TS 29.274 / 29.281	GTP-C v2 / GTP-U
3GPP TS 29.272 / 29.500 series	S6a Diameter / 5G SBI common and per-NF APIs
3GPP TS 32.251 / 32.255 / 32.260	Charging: packet-switched (4G) / 5G data connectivity / IMS
3GPP TS 33.401 / 33.501	4G / 5G security architecture (incl. SUCI)
3GPP TS 33.126 / 33.127 / 33.128	Lawful interception requirements / architecture / stage 3
GSMA IR.92 / IR.94 / NG.114	VoLTE / video / 5GS IMS profiles
GSMA IR.51	IMS over Wi-Fi profile